# Lab Values Cheat Sheet

Adult reference intervals in US conventional and SI units, conversion factors, nursing cues and unit traps for NCLEX review

MindShelf

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**TRAP Reference intervals vary between laboratories**

Interpret a result against the interval printed on **your lab's report**. The values below are one published set, for adults, for study.

## Adult reference intervals

| Test | US conventional | Factor | SI |
|---|---|---|---|
| **Electrolytes** | | | |
| Sodium ($Na^+$) | 136–145 mEq/L | × 1 | 136–145 mmol/L |
| Potassium ($K^+$) | 3.5–5.0 mEq/L | × 1 | 3.5–5.0 mmol/L |
| Chloride ($Cl^-$) | 98–106 mEq/L | × 1 | 98–106 mmol/L |
| Bicarbonate ($HCO_3^-$), serum | 23–28 mEq/L | × 1 | 23–28 mmol/L |
| Total $CO_2$, serum | 23–30 mEq/L | × 1 | 23–30 mmol/L |
| Calcium, total | 8.6–10.2 mg/dL | × 0.2495 | 2.15–2.54 mmol/L |
| Calcium, ionized | 4.65–5.25 mg/dL | × 0.2495 | 1.16–1.31 mmol/L |
| Magnesium | 1.6–2.6 mg/dL | × 0.4114 | 0.66–1.07 mmol/L |
| Phosphorus (as P) | 3.0–4.5 mg/dL | × 0.3229 | 0.97–1.45 mmol/L |
| **Glucose** | | | |
| Glucose, fasting plasma | 70–99 mg/dL | × 0.0555 | 3.9–5.5 mmol/L |
| Hemoglobin A1c | 4.0–5.6 % | IFCC eq. | 20–38 mmol/mol |
| **Kidney** | | | |
| BUN (urea nitrogen) → urea | 8–20 mg/dL | × 0.357 | 2.9–7.1 mmol/L |
| Creatinine, female | 0.50–1.10 mg/dL | × 88.4 | 44–97 µmol/L |
| Creatinine, male | 0.70–1.30 mg/dL | × 88.4 | 62–115 µmol/L |
| **Blood count** | | | |
| Hemoglobin, female | 12–16 g/dL | × 10 | 120–160 g/L |
| Hemoglobin, male | 14–18 g/dL | × 10 | 140–180 g/L |
| Hematocrit, female | 37–47 % | × 0.01 | 0.37–0.47 L/L |
| Hematocrit, male | 42–50 % | × 0.01 | 0.42–0.50 L/L |
| WBC (leukocytes) | 4,000–11,000 /µL | × 0.001 | 4.0–11.0 ×10⁹/L |
| Platelets | 150,000–450,000 /µL | × 0.001 | 150–450 ×10⁹/L |
| **Coagulation** | | | |
| PT (prothrombin time) | 11–13 s | same | 11–13 s |
| INR, no anticoagulant | 0.8–1.1 | same | 0.8–1.1 |
| INR target on warfarin | 2.0–3.0 | same | 2.0–3.0 |
| aPTT | 25–35 s | same | 25–35 s |
| **Arterial blood gas (room air)** | | | |
| pH | 7.38–7.44 | same | 7.38–7.44 |
| $PaCO_2$ | 38–42 mm Hg | × 0.1333 | 5.1–5.6 kPa |
| $PaO_2$ | 75–100 mm Hg | × 0.1333 | 10.0–13.3 kPa |
| $HCO_3^-$ | 23–26 mEq/L | × 1 | 23–26 mmol/L |
| $SaO_2$ | ≥ 95 % | × 0.01 | ≥ 0.95 (fraction) |
| **Liver panel** | | | |
| Albumin | 3.5–5.5 g/dL | × 10 | 35–55 g/L |
| Total protein | 5.5–9.0 g/dL | × 10 | 55–90 g/L |
| Bilirubin, total | 0.3–1.0 mg/dL | × 17.1 | 5–17 µmol/L |
| ALT and AST (each) | 10–40 U/L | × 0.0167 | 0.17–0.67 µkat/L |
| Alkaline phosphatase | 30–120 U/L | × 0.0167 | 0.50–2.00 µkat/L |
| **Lipids (decision limits)** | | | |
| Total cholesterol, desirable | < 200 mg/dL | × 0.0259 | < 5.18 mmol/L |


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| Test | US conventional | Factor | SI |
|---|---|---|---|
| LDL cholesterol, optimal | < 100 mg/dL | × 0.0259 | < 2.59 mmol/L |
| HDL, low (female) | < 50 mg/dL | × 0.0259 | < 1.30 mmol/L |
| HDL, low (male) | < 40 mg/dL | × 0.0259 | < 1.04 mmol/L |
| Triglycerides, fasting | < 150 mg/dL | × 0.0113 | < 1.70 mmol/L |
| **Drug level** | | | |
| Digoxin, general range | 0.5–1.9 ng/mL | × 1.28 | 0.6–2.4 nmol/L |

*SI = US × factor; the SI value is calculated from the US value, except ionized calcium (source in mmol/L). Blood gas values: room air.*

## Convert US ↔ SI

**SI = US × factor**; US = SI ÷ factor. mg/dL → mmol/L: factor = 10 ÷ molar mass (g/mol). mEq/L → mmol/L: ÷ ion charge ($Na^+$, $K^+$, $Cl^-$, $HCO_3^-$: same number). mm Hg → kPa: × 0.1333 (1 mm Hg ≈ 133.322 Pa). HbA1c: mmol/mol = (% − 2.15) × 10.929 (IFCC eq.).

**EXAMPLE Three conversions**
Glucose 180 mg/dL × 0.0555 = 9.99 ⇒ **10.0 mmol/L**.
Creatinine 1.5 mg/dL × 88.4 = 132.6 ⇒ **133 µmol/L**.
Ionized Ca 1.1 mmol/L ÷ 0.2495 ≈ 4.4088 ⇒ **4.41 mg/dL**.

## Unit traps

**TRAP Same test, two numbers**
Glucose 99 mg/dL = 5.5 mmol/L. Read the unit before you compare a result with a range.

**TRAP mEq is not always mmol**
Equal only for single-charge ions. $Ca^{2+}$ and $Mg^{2+}$: 1 mmol = 2 mEq.

**TRAP BUN is not urea**
US labs report urea **nitrogen** (BUN); SI labs report urea. Urea has 2 N atoms: × 0.357.

**TRAP Total vs ionized calcium**
Low albumin lowers **total** calcium; ionized calcium (the active fraction) may be normal. Check which one the order and the report give.

**TRAP Hemolyzed sample**
Red-cell breakdown in the tube releases potassium: a falsely high $K^+$. Unexpected high $K^+$: ask whether the sample was hemolyzed and a redraw is needed.

**TRAP Hematocrit and $SaO_2$ in SI**
Written as fractions: 37% = 0.37 L/L; 95% = 0.95.

## Nursing cues

**CLINICAL NOTE INR and bleeding**
On warfarin, the usual INR target is 2.0–3.0. Above 3.0: higher bleeding risk. Report per protocol.

**CLINICAL NOTE Potassium and the heart**
$K^+$ outside the interval, high or low, can disturb cardiac rhythm: report promptly; ECG monitoring per protocol.

**CLINICAL NOTE Digoxin level**
The interval shown is a general one; the right level depends on the patient and the indication. Follow the prescriber's target.

*Sources: ABIM Laboratory Test Reference Ranges (January 2026); MedlinePlus (INR, digoxin). Factors calculated from IUPAC atomic weights and SI definitions.*

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